

Example 4 : Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end (see Fig. 12.9). The height of the cylinder is 1.45 m and its radius is 30 cm. Find the total surface area of the bird-bath. (Take $\pi = \frac{22}{7}$)

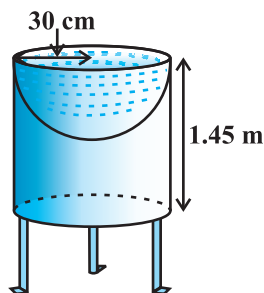


Fig. 12.9

Solution : Let h be height of the cylinder, and r the common radius of the cylinder and hemisphere. Then,

$$\begin{aligned}
 \text{the total surface area of the bird-bath} &= \text{CSA of cylinder} + \text{CSA of hemisphere} \\
 &= 2\pi rh + 2\pi r^2 = 2\pi r(h + r) \\
 &= 2 \times \frac{22}{7} \times 30(145 + 30) \text{ cm}^2 \\
 &= 33000 \text{ cm}^2 = 3.3 \text{ m}^2
 \end{aligned}$$

EXERCISE 12.1

Unless stated otherwise, take $\pi = \frac{22}{7}$.

- 2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
- A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.
- A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.
- A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.
- A hemispherical depression is cut out from one face of a cubical wooden block such that the diameter l of the hemisphere is equal to the edge of the cube. Determine the surface area of the remaining solid.
- A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends (see Fig. 12.10). The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

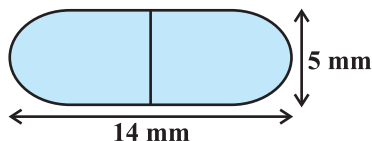


Fig. 12.10

Example 7 : A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Determine the volume of the toy. If a right circular cylinder circumscribes the toy, find the difference of the volumes of the cylinder and the toy. (Take $\pi = 3.14$)

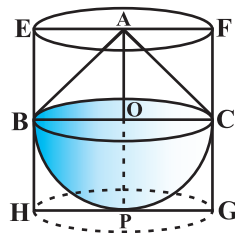


Fig. 12.14

Solution : Let BPC be the hemisphere and ABC be the cone standing on the base of the hemisphere (see Fig. 12.14). The radius BO of the hemisphere (as well as of the cone) = $\frac{1}{2} \times 4 \text{ cm} = 2 \text{ cm}$.

$$\begin{aligned} \text{So, volume of the toy} &= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h \\ &= \left[\frac{2}{3} \times 3.14 \times (2)^3 + \frac{1}{3} \times 3.14 \times (2)^2 \times 2 \right] \text{ cm}^3 = 25.12 \text{ cm}^3 \end{aligned}$$

Now, let the right circular cylinder EFGH circumscribe the given solid. The radius of the base of the right circular cylinder = HP = BO = 2 cm, and its height is

$$EH = AO + OP = (2 + 2) \text{ cm} = 4 \text{ cm}$$

$$\begin{aligned} \text{So, the volume required} &= \text{volume of the right circular cylinder} - \text{volume of the toy} \\ &= (3.14 \times 2^2 \times 4 - 25.12) \text{ cm}^3 \\ &= 25.12 \text{ cm}^3 \end{aligned}$$

Hence, the required difference of the two volumes = 25.12 cm^3 .

EXERCISE 12.2

Unless stated otherwise, take $\pi = \frac{22}{7}$.

1. A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π .
2. Rachel, an engineering student, was asked to make a model shaped like a cylinder with two cones attached at its two ends by using a thin aluminium sheet. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of air contained in the model that Rachel made. (Assume the outer and inner dimensions of the model to be nearly the same.)